

MITRE ATT&CK; Framework Guide for AI Models

This document provides a structured explanation of the MITRE ATT&CK; framework to help train AI models to classify cyber attack events into stages such as Initial Access, Execution, Lateral Movement, and Impact.

1. What is MITRE ATT&CK;?

MITRE ATT&CK; is a knowledge base of adversarial tactics and techniques based on real-world observations. It is used to understand how attackers behave and how to detect them.

2. Structure of the Matrix

The matrix consists of:

- **Tactics:** High-level goals (e.g., Initial Access, Execution, Impact)
- **Techniques:** Methods used to achieve those goals
- **Sub-techniques:** More detailed breakdowns of techniques

3. Common Tactics

Tactic	Description
Initial Access	How attacker enters system
Execution	Running malicious code
Persistence	Maintaining access
Lateral Movement	Moving across systems
Command & Control	Communicating with attacker
Impact	Final damage or disruption

4. Example Attack Mapping

- USB Inserted → Initial Access (T1200)
- File Executed → Execution (T1059)
- SCADA Connection → Lateral Movement (T1021)
- PLC Command → Impact (T0830)

5. AI Model Usage

The AI model should take log events as input and output: tactic, technique, and MITRE ID.
Example: USB inserted → Initial Access, Hardware Additions, T1200.